Bo Zhang

Allen, TX 75002 | 469-732-0166 | bozhang.bio@gmail.com

LinkedIn: https://www.linkedin.com/in/bozhangdata/ | GitHub: theophiluszhang.github.io/Bo_Portfolio

SUMMARY

Data Analyst / Data Engineer with 3 + years of experience building healthcare analytics pipelines, predictive models, and scalable reporting solutions using Python, SQL. Experienced in designing production-grade ETL workflows with data quality validation, schema alignment, API integrations, and automated pipeline orchestration. Strong background in healthcare data domains including claims, revenue cycle, staffing, and operational analytics. Familiar with Apache Spark and distributed data concepts through graduate coursework, currently expanding expertise in Azure ML, Databricks, and modern cloud data engineering practices.

TECHNICAL SKILLS

- **Programming & Data Engineering:** Python (pandas, NumPy, scikit-learn), SQL/T-SQL, Spark fundamentals, ETL pipeline development, API integrations, data validation, schema alignment, workflow automation
- **Cloud & Data Platforms:** Azure SQL Server, Power BI, Git/GitHub, Azure ML, Jupyter Notebook
- **Analytics & Modeling:** Power BI semantic modeling, feature engineering, model validation, Machine learning, statistical analysis
- **Healthcare & Business Domains:** revenue cycle management, staffing analytics, healthcare operational data
- **Tools & Platforms:** Alteryx, VS Code, AI-assisted development(Codex), Power Apps, Power Automate

PROFESSIONAL EXPERIENCE

PNC Bank - Data Engineer, Associate **August 2026 - Present6**

- Develop PySpark data transformation pipelines based on source-to-target mappings.
- Build and configure Apache Oozie workflows to orchestrate data processing jobs and manage dependencies.

BHG Recovery - Data Analyst/ Data Engineer **August 2024 - April 2026**

- Built and maintained a dozen automated ETL pipelines using Python and SQL to integrate vendor APIs, semi-structured healthcare data, and operational systems into Azure SQL Server. Used Python for API extraction, JSON processing, transformation, error handling, and automation, and SQL for schema alignment, de-duplication, validation, optimization, and idempotent upserts. Projects included Zoom API integration for telehealth monitoring and CredentialStream API integration for provider licensing, credentialing, compliance, and operational reporting.
- Developed a healthcare claims-denial prediction model in Python using pandas, scikit-learn, CatBoost, and XGBoost on 3M+ claims. Applied leakage-safe feature engineering and model benchmarking across logistic regression, random forest, CatBoost, and XGBoost, achieving a PR-AUC of 0.973 on a 200K holdout set and delivering pilot scores aligned with revenue-cycle capacity.
- Designed scalable SQL reporting datasets and Power BI semantic models supporting staffing, IT operations, finance/RCM, and revenue cycle analytics across multiple business stakeholders.

Integrated Silicon Solution, Inc. - Data Analyst **June 2023 - August 2024**

- Designed forecasting and supply-chain analyses (sales, backlog, inventory) to improve planning and cost efficiency; developed dashboards and metrics using SQL and Power BI to monitor supply-chain health for operations.

EDUCATION

Boston University - M.S., Computer Science and AI (Expected Dec 2028)

The University of Texas at Dallas - M.S., Business Analytics (May 2023)

Relevant coursework: Applied ML, Predictive Analytics, Statistics for DS, Big Data, Databases, Applied NLP